

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H1



**ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS**

H1 BIOLOGY

8876/02

STRUCTURED QNS AND ESSAY (Mark Scheme)

Monday

30th August 2021

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagram, graph.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Write your answers in the spaces provided on the question paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' Use	
STQ	
1	/10
2	/20
3	/15
4 or 5	/15
Total	/60

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

This document consists of **xx** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

1(a) T-helper lymphocytes and Leydig cells are two types of mammalian cells. The main role of T-helper lymphocytes and Leydig cells is to synthesise and secrete cell-signalling molecules (which are molecules that are released by one cell. Such cell-signalling molecules, e.g cytokines and testosterone, recognise and bind to receptor on the cell surface membrane of another cell where they cause a cellular response).

- T-helper lymphocytes synthesise proteins known as cytokines.
- Leydig cells synthesise the steroid (lipid) hormone testosterone from cholesterol. Leydig cells also synthesise cholesterol.

Figure 1.1 shows part of a mammalian cell.

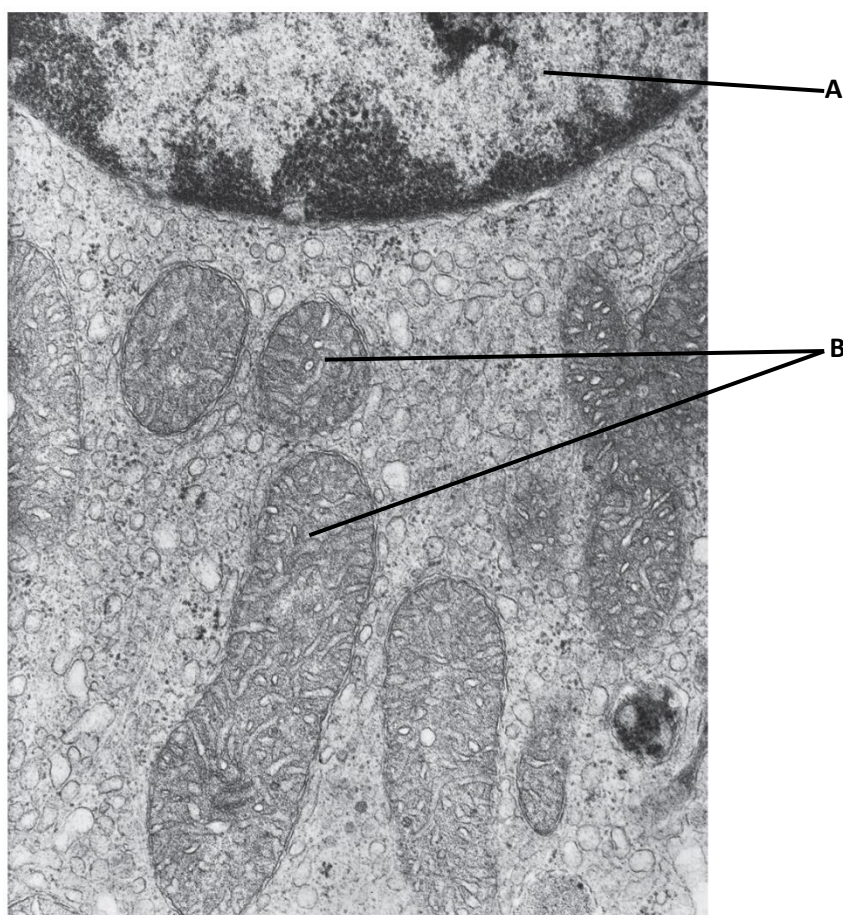


Fig. 1.1

(i) Identify organelles **A** and **B** on Fig. 1.1.

A:

B:

[1]

A: nucleus [reject: chromatin, nucleolus]

B: mitochondria [reject: mitochondrion]

- (ii) Organelle **A** is involved in mitotic cell cycle. State the process that occurs during interphase to produce genetically identical daughter cells.

.....[1]

1 Semi-conservative DNA replication ;

- (iii) Explain whether Fig. 1.1 shows part of a Leydig cell or part of a T-helper lymphocyte.

.....

[2]

1. Leydig cell ;
2. much / many / high proportion of, smooth endoplasmic reticulum ; (sER present) that functions to synthesise, cholesterol / testosterone / steroid hormones / lipids ;

OR

no / little, rough endoplasmic reticulum ; (rER present) that function to synthesise polypeptide at ribosomes on its surface / chemical modification of protein in the lumen of rER ;

- (b) Cell-surface receptors are membrane-anchored (integral) proteins that bind to external signalling molecules.

Such receptors are synthesised within the cell and brought to the cell surface membrane as shown in the Figure 1.2.

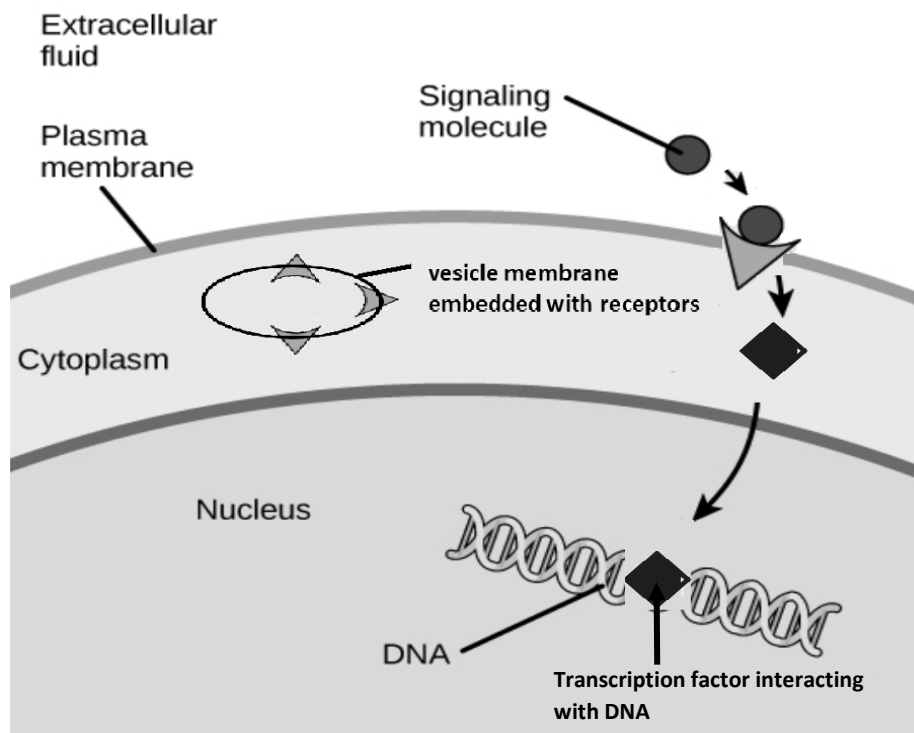


Fig. 1.2

With reference to Fig. 1.2, describe how the receptor is brought to and become part of the cell surface membrane.

.....

[2]

- 1 Secretory vesicles **with embedded receptors** (with ligand-binding site of the receptor projecting into the vesicle) **move towards** the cell surface membrane (of cells);
- 2 Vesicle's membrane fuses with the **cell surface membrane** ; resulting in the **ligand-binding site** of the receptor **projecting into the extracellular fluid**
- 3 Ref. ATP expenditure for **movement** of vesicles on the **microtubules / filaments of the cytoskeleton** ;

- (c) *Eichhornia crassipes*, a water hyacinth plant, is an invasive species found in lakes and rivers throughout Asia. They grow rapidly, producing many flowers and seeds. Each flower of this plant has six anthers (each anther is at one end of the filament) and one stigma as shown in Figure 1.3.

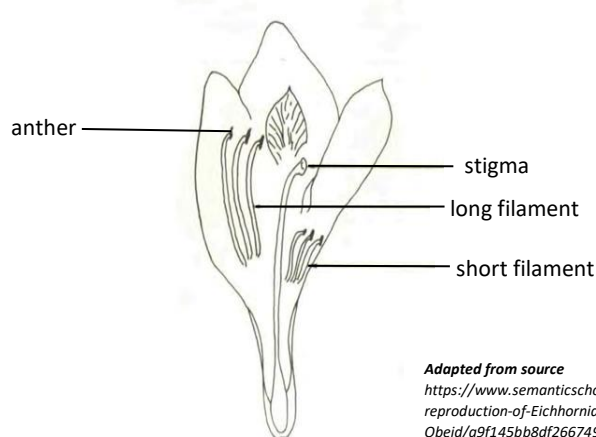


Fig. 1.3 showing spatial arrangement of stamen (comprising of both anther and filament) and stigma in *E. crassipes*.

Each anther was observed to release many pollen grains. After pollination, the pollen grains germinate after it comes into contact with the stigma, in response to the sugary fluid secreted by the mature stigma. A pollen tube grows out from each pollen grain. The pollen tube serves to transport the male gamete cells to the female gamete cells (ovules) in the flower.

The tip of the pollen tube is protected by an outer layer of pectin.

Pectin is a polysaccharide.

Starch grains, mitochondria and vesicles are present in the tip of the pollen tube.

Using the information given and your content knowledge on organelles and carbohydrates, describe how the production of pectin occurred at the tip of a pollen tube.

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.....[4]

1. Starch is {digested / broken down / hydrolysed} to release glucose ;
2. Condensation reactions with release of 1 water molecule per glycosidic bonds are used to form pectin ;
3. Pectin {secreted into / enclosed in / transport in} vesicles ;

4. Vesicles **fuse** with the {tip / cell surface membrane} to **release** {pectin / content} ; [**ACCEPT**: reference to exocytosis releasing contents]
5. Cellular respiration in mitochondria releases {ATP / ~~energy~~} for production of {vesicles / pectin / enzymes / other relevant molecules} ;
[**ACCEPT**: energy used for movement of vesicles in exocytosis]

[Total: 10]

2(a) Severe combined immunodeficiency (SCID) is a disease that affect the development and maturation of white blood cells known as lymphocytes.

There are two general types of SCID, both of which are caused by recessive alleles.

Fig. 2.1 shows a family pedigree including some people with a type of SCID.

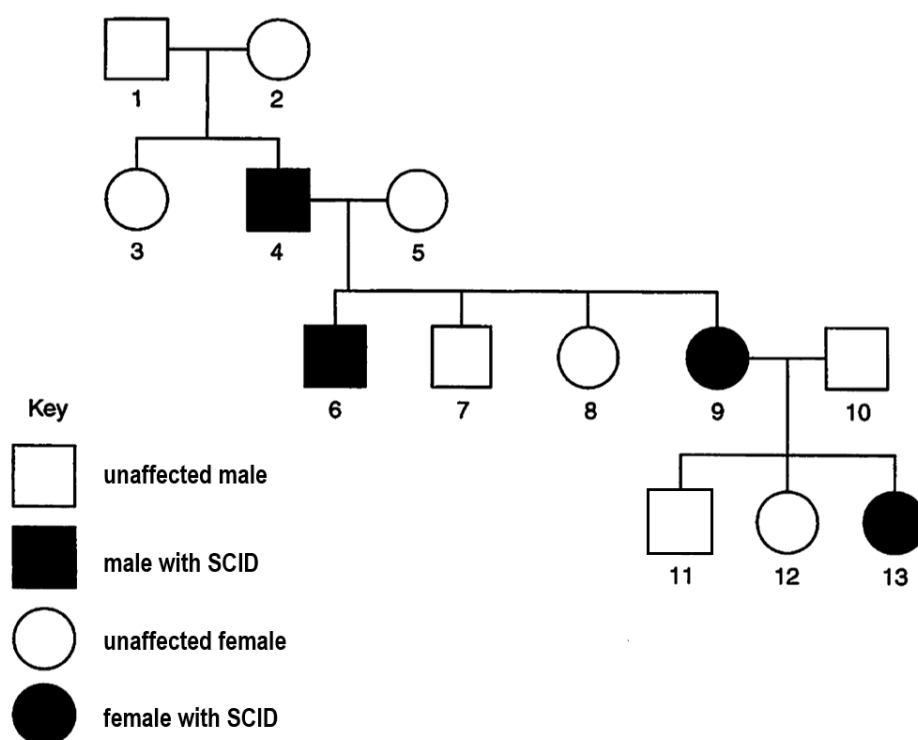


Fig. 2.1

(i) Explain what is meant by the term allele.

.....

.....

.....

.....[2]

- 1 Allele: **alternative form** of a gene that occupy the same gene locus on homologous chromosomes;
- 2 causes **slightly different phenotypes** for the same characteristic due to a **slight difference in DNA sequence**;

(ii) Explain, with evidence in Fig. 2.1, whether the disease is X-linked.

.....

[3]

- 1 Not X-linked / autosomal;
- 2 If X linked recessive, all diseased mother, e.g. individual 9, will have diseased sons but individual 11 is not diseased;
 / Individual 2 is not diseased but have a diseased son individual 4;
 / equal number of males, individuals 4,6, and females individuals 9,13 affected;
- 3 Ref. son of diseased mother will inherit X chromosome containing disease allele from mother;

(b) In mice, the intensity of pigmentation of the fur is controlled by multiple alleles of a single gene.

The alleles are listed below in order of dominance, with **C** as the most dominant.

- **C** = full colour
- **C^{ch}** = chinchilla
- **C^h** = himalayan
- **C^p** = platinum
- **C^a** = albino

Eye colour in mice is controlled by two alleles of a single gene, **B/b**:

- allele **B** codes for black eyes
- allele **b** codes for red eyes.

A mouse with full colour fur and black eyes was crossed with a mouse with himalayan fur and black eyes. One of the offspring was albino with red eyes.

Using the symbols above, draw a genetic diagram to show the genotypes and phenotypes of the offspring of this cross.

Parental phenotype	full colour fur and black eyes mouse	x	himalayan fur and black eyes mouse
Parental genotype	CC^aBb	x	C^hC^aBb
Parental gametes	$\textcircled{CB}$ $\textcircled{Cb}$ $\textcircled{C^aB}$ $\textcircled{C^ab}$	x	$\textcircled{C^hB}$ $\textcircled{C^hb}$ $\textcircled{C^aB}$ $\textcircled{C^ab}$

Punnett square

	$\textcircled{C^hB}$	$\textcircled{C^hb}$	$\textcircled{C^aB}$	$\textcircled{C^ab}$
$\textcircled{CB}$	CC^hBB full black	CC^aBB full black	CC^hBb full black	CC^aBb full black
$\textcircled{Cb}$	CC^hBb full black	CC^aBb full black	CC^hbb full red	CC^abb full red
$\textcircled{C^aB}$	C^aC^hBB Him black	C^aC^aBB albino black	C^aC^hBb Him black	C^aC^aBb albino black
$\textcircled{C^ab}$	C^aC^hBb Him black	C^aC^aBb albino black	C^aC^hbb Him red	C^aC^abb albino red

F₁ phenotypic ratio

6 full black : 3 himalayan black : 3 albino black : 2 full red : 1 himalayan red : 1 albino red

Mark scheme:

1. Parental genotypes ;
2. Parental gametes (**circled**) ;
3. F1 genotypes / Punnett square ;
4. F1 genotypes correspond to phenotypes ;
5. F1 phenotypic ratio

(c) Figure 2.2 shows the nematode worm, *Caenorhabditis elegans*.



Fig. 2.2

Stem cells of *C. elegans* have been studied.

Figure 2.3 shows the change in mass of DNA per nucleus in a stem cell during one cell cycle. $1\text{pg} = 1 \times 10^{-12}\text{g}$.

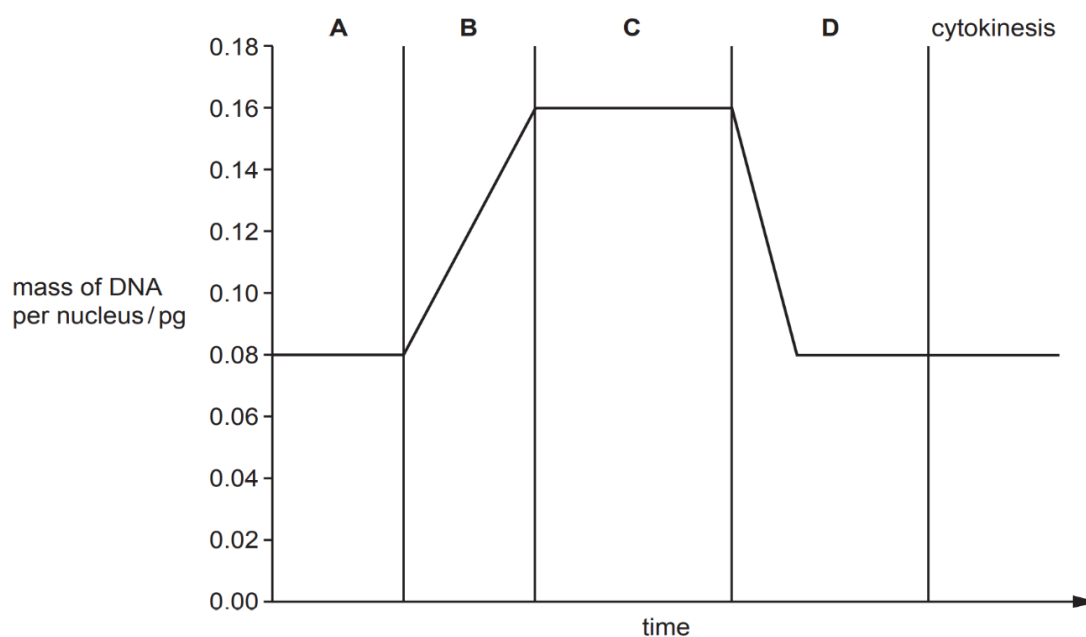


Fig. 2.3

(i) State the phases of interphase shown by **A** and **B** and the stage of mitosis shown by **D** in Fig. 2.3.

A

B

D

1 A = G1 (phase)

2 B = S (phase)

3 D = telophase

[Ignore G2 phase]

[1]

(ii) Outline what happens in a cell in stage D in preparation for cytokinesis.

.....

[2]

Any two:

- 1 (re)formation of nuclear envelope ; nuclear envelope, forms around each group of chromosomes ;
- 2 Chromosomes de-condensing into chromatin;
- 3 (movement of) organelles to be shared between two daughter cells ;
- 4 Spindle, disassemble / break down / degrades / disappear / AW ;
- 5 Cleavage furrow forms ; cell surface membrane, pinches in / constricts / infolds ;

(iii) Immediately after cytokinesis, daughter cells are **not** identical even though they are genetically identical.

Suggest a reason why daughter cells are not identical immediately after cytokinesis.

.....

[1]

- 1 Unequal sharing out of, cytoplasm / organelles / named organelles, (at cytokinesis) ;
 [ACCEPT: unequal sizes]
 [Reject: plasmids / circular DNA in mitochondria, as focus of question is not on genetic factors]

(iv) A stem cell of a female *C. elegans* has 12 chromosomes.

Complete Table 2.1 to show the number of nuclei within the stem cell in stages **A**, **B** and **D** of the cell cycle shown in Figure 2.3 and the number of chromosomes in each nucleus during these stages.

Table 2.1

ANS:

	stage of the cell cycle		
	A	B	D
number of nuclei within the stem cell	1	1	2 ;
number of chromosomes in each nucleus	12	12	12

1 1 mark each for correct answer for each row

[2]

- (d) Embryonic stem cells are extracted from human embryos 4-5 days after fertilisation, when the embryo consists of 50 to 150 cells.

Suggest one ethical concern about the use of embryonic stem cells.

.....
[1]

[Specific ethical issues associated with the use of ES cells]

- 1 Destruction of embryos donated by individuals undergoing IVF as they are **potential human lives** ;
 - 2 Intentionally creating and destroying embryos for via SCNT **violates respect** for human **lives** ;
 - 3 **Medical risks** of oocyte retrieval from the ovaries for SCNT as it is an invasive procedure ;
 - 4 **Reproductive interest** of women undergoing fertility treatment may be compromised if they consent to donate any of their oocytes for research purposes
- [Any 1]

- (e) The tomato plant, *Solanum lycopersicum*, does not tolerate periods of drought (water shortage). Researchers have produced a tomato plant that has an improved tolerance of drought.

The nuclei of plants produce small lengths of RNA known as microRNAs.

MicroRNAs in guard cells have been shown to prevent the synthesis of some proteins.

The guard cells of the drought-tolerant tomato plants produced more microRNA molecules than the guard cells of the non-tolerant plants.

MicroRNA molecules do **not** prevent transcription but interact with messenger RNA (mRNA).

Suggest how this microRNA can interact with mRNA to prevent the production of proteins in guard cells of *S. lycopersicum*.

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.....[3]

1. microRNA {**binds** to mRNA / forms hydrogen bonds with (bases on) mRNA}
2. bases in microRNA are complementary to bases on mRNA ; [Accept: binds by complementary base pairing]
3. mRNA with microRNA **cannot bind** to (small subunit of) ribosome ;
/ anticodon of tRNA cannot bind to (some) codons on mRNA ;
4. AVP ; e.g. complex of microRNA and mRNA recognised for **degrading** ;

[Total: 20]

3(a) ATP is often known as the universal 'energy currency' of the cell.

Most ATP are made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Figure 3.1 shows two of such membrane systems.

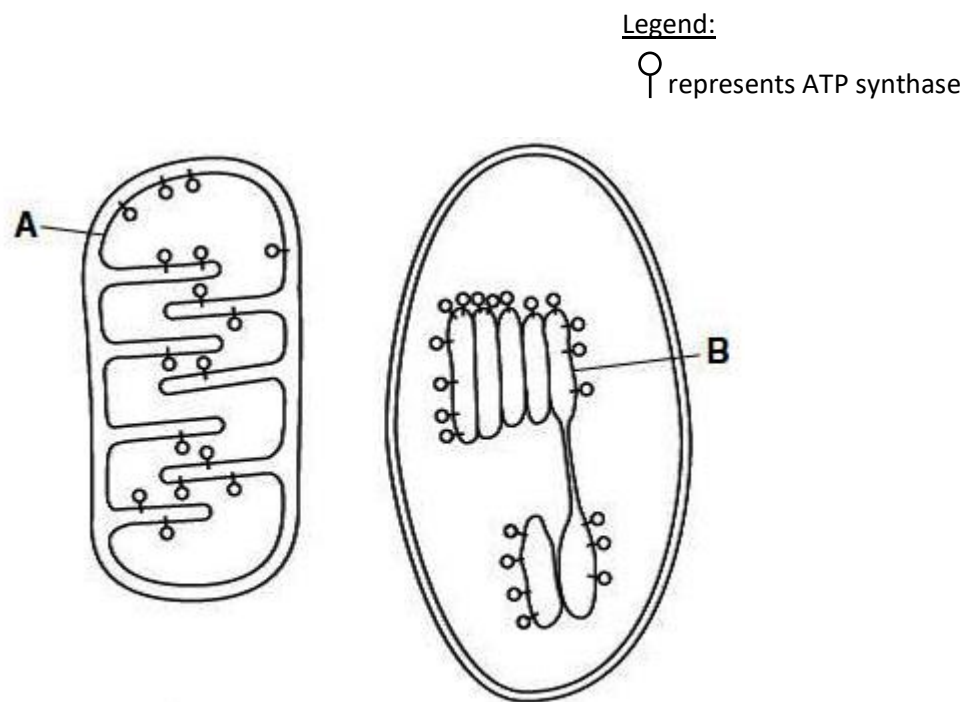
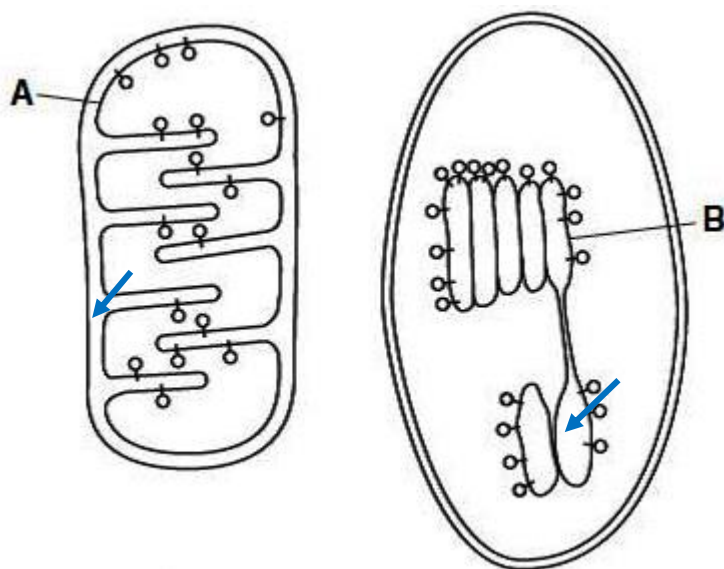


Fig. 3.1

- (i) Draw arrows onto **each** of the membrane systems in Fig. 3.1 to show the direction in which protons are pumped.

.....[1]

- 1** A – from matrix into intermembrane space, B – from stroma into thylakoid space;
 [Reject: if arrow goes through ATP synthase]



- (ii) Outline how energy is generated for the pumping of protons across membrane system B.

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.....[3]

- 1 [ETC] Electrons captured by {primary electron acceptor / **special chlorophyll a pigments** [REJECT: NADP]} are passed down electron carriers in electron transport chain (ETC) ;
- 2 electron carriers are arranged in progressively **decreasing energy levels** ;
- 3 **Energy released** by **transfer of electrons** / **electron flow** is used to pump H^+ from stroma to thylakoid space.

- (iii) Explain why transmembrane proteins are necessary for protons to be pumped across the membrane systems.

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.....[2]

- 1 Protons are charged ions ;
- 2 They are therefore **unable to pass through** the **hydrophobic core** of the membranes ;

- (iv) It is also known that there are cyclic pathways involving the use of enzymes occurring in both membrane systems.

State the names of the two cyclic pathways and outline the differences between them.

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.....[4]

1 State pathways	<u>Calvin cycle</u>	<u>Krebs cycle</u>
Differences		
2 Location	<u>Stroma: of chloroplast</u>	<u>Mitochondrial matrix: of mitochondria</u>
3 Electron/hydrogen carriers involved	<u>NADP⁺ / NADPH</u>	<u>NAD⁺/NADH, FAD/FADH₂</u>
4 Carbon dioxide used/released	CO₂ enters the Calvin cycle, and is fixed by combining with a 5C compound called ribulose biphosphate (RuBP) , catalysed by the enzyme <u>rubisco</u> (RuBP carboxylase)	CO₂ is released during oxidative decarboxylation
5 ATP used/synthesized	Used in reduction phase when PGA is reduced to PGAL ; and used in regeneration of RuBP from PGAL	Synthesised by substrate level phosphorylation [Reject: oxidative phosphorylation]

- (b) Sehuencas water frogs are endemic to Bolivia. It was once believed that this frog was extinct in the wild. This was due to human activity and a disease caused by a fungus.

In 2008, there was only one Sehuencas water frog in captivity.

In 2018, five Sehuencas water frogs were discovered in a Bolivian mountain forest.

Three of these frogs were male. Scientists suggested that these five frogs were resistant to the fungal disease.

Explain how resistance to this fungus came about in a population of Sehuencas water frogs.

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.....[5]

- 1 Genetic variation exist in the (Sehuencas water frogs) population due to spontaneous mutation ;
- 2 **Fungus causing disease in the frogs** acts as selection pressure ;
- 3 Frogs with (new) **allele** (coding for protein) **involved in fungal resistance** are at selective advantage ;
- 4 Frogs resistant to the fungus survive and produce viable and fertile offspring ;
Allele for fungus resistance are **passed down to their offspring** ;
- 5 Over time, allele frequency change. **Frequency of allele in fungus resistance increases** and become predominate in the population.

[Total: 15]

Section B

Answer EITHER 4 or 5.

Write your answers on lines provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

- 4 (a)** Outline the process of DNA replication. [5]
- (b)** Explain how different types of gene mutation can affect the phenotype and outline the effects of the mutant alleles that cause sickle cell anemia on the phenotype of a person. [10]

[Total: 15]

- 5 (a)** Many different polymers are formed in cells. A polymer is a chain of monomers. [8]
- Describe the polymerisation of different types of biomolecules in a plant.
- (b)** Discuss how the complementary shapes between (bio)molecules enable eukaryotic cells to carry out their diverse processes. [7]

[Total: 15]

Essay 4

- (a)** Outline the process of DNA replication. [5]

- 1 DNA helicase unwind and unzip by breaking hydrogen bonds between complementary base-pairs at origin of replication (ori).
- 2 Both DNA strands act as templates for DNA replication.
- 3 DNA primase attached to each DNA template strand to synthesise RNA primer molecules which are attached to each strand (at origin of replication).
- 4 RNA primer molecules have base sequences complementary to the base sequences of DNA template strand ;
- 5 DNA polymerase adds nucleotides to 3' OH ends of primers and synthesises new strands of DNA in a 5' to 3'direction.

- 6 One strand, called the leading strand, is synthesised in continuous long sections.
The other strand, called the lagging strand, is synthesised in short sections.
- 7 DNA polymerase hydrolyses the RNA primers and the gaps (left by hydrolysed RNA primers) are replaced with DNA nucleotides on both strands.
- 8 DNA ligase catalysed the phosphodiester bonds between Okazaki fragments to form the (continuous sequence in) lagging strand.

4(b) Explain how different types of gene mutation can affect the phenotype and outline the effects of the mutant alleles that cause sickle cell anemia on the phenotype of a person. [10]

gene mutation:

- 1 base substitution ;
- 2 silent mutation / neutral mutation / missense mutation ; (often) does not have a significant effect on phenotype ; OR
neutral mutation ; (often) does not have a significant effect on phenotype ; OR
missense mutation ; may have a significant effect on phenotype e.g sickle cell anemia ;
- 3 base, insertion / deletion lead to,
- 4 frameshift ; (so) has significant effect on phenotype ;
- 5 change in, primary structure / amino acid sequence / polypeptide made ;
change in, tertiary structure / 3D conformation ;
- 6 loss of function in protein or enzyme / example described ;
- 7 (premature) stop codon ; result in truncated protein or enzyme (that are non-functional) ;
- 8 Ref. central dogma ;

sickle cell anaemia:

- 9 mutant allele (HbS) is co-dominant (to normal allele, HbA) ;
- 10 homozygote will have sickle cell anaemia / heterozygote will have sickle cell trait ;
- 11 (gene mutation will result in) hydrophobic valine replaced by hydrophilic glutamic acid in the polypeptide ;
- 12 Change in the chemical property of the R group of the amino acid (when valine replaces glutamic acid at the sixth amino acid position) changes the **interaction** between R-groups of amino acids, thus affecting protein folding to form a **different tertiary structure (3-D conformation)** for HbS ;
- 13 [Effect on HbS occurs] at low oxygen concentration, **solubility** of HbS **decreases** which leads to ;
- 14 HbS molecules stick to each other via their **hydrophobic** regions and polymerise into fibres. [**Reject: HbS sickles**]
- 15 Affected red blood cells (**REJECT: HbS**) will adopt a sickle shape ;
- 16 Due to their shape, there is tendency of the sickled red blood cells to **clump together** and **clog capillaries**, preventing other cells from moving through capillaries ;

- 17 **Obstructing blood flow** to organs can cause **organ damage** (as a result of oxygen in RBCs not being able to reach the organs) ;
- 18 Sickled shaped RBCs have **shorter lifespan**, resulting in anaemia in patients.

Essay 5

(a) Many different polymers are formed in cells. A polymer is a chain of monomers.

Describe the polymerisation of different types of biomolecules in a plant. [8]

1. Polymerisation of all the biomolecules occur through condensation ; water is **lost** / **removed**

[Starch]

2. formed from many α -glucose monomers linked together via α -1,4 glycosidic bonds
3. forming amylose and amylopectin
4. A α -1,4 glycosidic bond is formed between the OH group of carbon 1 of an α - glucose and OH group on carbon 4 of the **other** α -glucose (do not double award in cellulose)
/ A α -1,6 glycosidic bond is formed between the OH group of carbon 1 of an α - glucose and OH group on carbon 6 of the **other** α -glucose
5. amylopectin is a branched polymer where the branch points are linked by α -1,6 glycosidic bonds
6. results in a **helical** chain / polymer coiled into a helical shape

[Cellulose]

7. formed from many β -glucose monomers linked together via β -1,4 glyosidic bonds
8. A β -1,4 glycosidic bond is formed between OH group of carbon 1 of an β -glucose and OH group of carbon 4 of the **other** β -glucose (do not double award in starch)
9. **Alternate/successive/neighbouring** glucose monomers are being **inverted/rotated 180° relative to one another**
10. Resulting in a polymer which is **straight** / straight chain
11. catalysed by an enzyme known as cellulose synthase (located on the cell membrane of plant cells)

[Protein]

12. formed from many amino acids linked together via peptide bonds
13. A peptide bond is formed between the carboxylic acid group / (-COOH) of one amino acid and the **amino group / (-NH₂)** of **another** amino acid
14. Catalyzed by peptidyl transferase (of ribosomes) ; during translation

[Nucleic acid] / [DNA and RNA]

15. formed from many **nucleotides** / nucleoside triphosphates linked together via phosphodiester bonds
16. A phosphodiester bond is between the 3'-OH group of one nucleotide and the 5' phosphate group of an **incoming** / **another** nucleotide
17. Catalyzed by RNA polymerase ; during transcription
OR DNA polymerase ; during DNA replication

5(b) Discuss how the complementary shapes between biomolecules enable eukaryotic cells to carry out their diverse processes. [7]

[Enzyme-substrate]

1. Substrates have **complementary 3D conformation** to enzyme's active site
2. To form **enzyme-substrate complex**; results in **lowering of activation energy**
3. **Product** is formed, resulting in a **different 3D conformation** / no longer complementary to active site ; hence **product is released**, no longer remain bound at active site
4. Max 1 example on enzyme-substrate interaction
 5. [ELK: Lock and Key] Lock and key hypothesis → Shape of active site is exactly complementary to shape of substrate
 6. [EAS: Active site] Shape of active site is complementary to shape of substrate → Bind to form ES complex → Catalytic function (Catabolic / Anabolic) → Product formation
 7. [EIF: Induced Fit] Induced fit hypothesis → Shape of active site is complementary to shape of substrate, but slight conformational change to fit more snugly
[NOTE: Phrase the above statement very carefully! The substrate and enzyme are complementary in shape.]
 8. [EE: Enzyme Example] Named example of enzyme (e.g. maltase binds and hydrolyzes maltose to α-glucose)
 9. [EP: Product] Once product is formed → Different shape → No longer remain bound at active site

[Photosynthesis / Respiration]

[Max 2]

1. [EP / ER: Photosynthesis / Respiration] Shape of ATP synthase is complementary to shape of ADP and P_i → Synthesize ATP → Drive metabolic reaction
2. [EP: Photosynthesis] Shape of NADP reductase is complementary to shape of $NADP^+$ and H^+ → Synthesize NADPH → Carbon reduction in Calvin cycle
[Accept: Other enzymes with substrates and functions stated]

[Transport across membrane]

10. Carrier protein has a **binding site complementary** in shape to **molecule to be transported**;
11. Binding of molecule to be transported to carrier protein induces a **conformational change** in carrier protein;
12. This allows for molecule to be **transported across** cell membrane;
13. Receptor-mediated endocytosis ; **binding site** of receptor at cell membrane is complementary to shape of molecule to be taken into cell

[Proteins-nucleic acid interaction]

14. Proteins may have DNA binding {domains / sites} that are **complementary** in shape to **specific DNA sequence**

[DNA replication]

15. Example: DNA helicase binds to origin of replication ; **initiate DNA replication** at the correct position
16. [PRep: Replication] Shape of DNA binding domain of helicase is complementary to shape of DNA sequence at Ori → Separate two parental strands
17. [PRep: Replication] Shape of DNA binding domain of primase / DNA polymerase / ligase is complementary to shape of 5' phosphate group of free (DNA) nucleotide and 3' OH group of adjacent nucleotide, as well as parental strand → Synthesize primers / Formation of phosphodiester bond between nucleotides of daughter strand / Seals nick between DNA fragments during replication
[Accept: Proofreading ability of DNA polymerase]

[Transcription]

18. Max 1 example:
1. E.g. RNA polymerase binds to promoter sequence ; **initiating transcription** at correct position
 2. [Transcription Factors] Shape of DNA binding domain of general transcription factor is complementary to shape of DNA sequence at promoter → allows transcription
 3. [Transcription] Shape of DNA binding domain of RNA polymerase is complementary to shape of 5' phosphate group of free RNA nucleotide and 3' OH group of adjacent nucleotide, as well as template strand → Formation of phosphodiester bond between nucleotides of mRNA during transcription

[Translation]

19. rRNA in large ribosomal subunit binds to tRNA via complementary shape
20. [Amino acid activation] Shape of amino acid + ATP + tRNA is complementary to shape of active site of aminoacyl-tRNA synthetase → Amino acid activation
21. [Peptidyl transferase] Shape of peptidyl transferase is complementary to shape of aminoacyl-tRNA → Formation of peptide bond between amino acids to form polypeptide during translation
22. [Release factor] Shape of release factor is complementary to shape of mRNA sequence at stop codon → Termination of translation

[Ribosomes]

23. **Large and small ribosomal subunits** are **complementary** in shape ; binds together to form ribosomes

[Haemoglobin]

24. Haemoglobin is made up of 4 polypeptides and their haem groups to form a specific conformation
25. Reference to cooperative binding between 4 polypeptides ; to **facilitate efficient binding to oxygen**

[Cell-cell adhesion]

26. Membrane **proteins / glycoproteins / glycolipids** of **adjacent cells** may be **attached together** via complementary shape ; for **cell-cell adhesion**